

**LAPORAN TUGAS PRAKTIK KOMPREHENSIF - STRUKTUR
SISTEM OPERASI**



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**PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS TEKNIK
UNIVERSITAS MUHAMMADIYAH JEMBER**

2025

Tugas Praktik Komprehensif

Tugas 1: Process Manager

```
GNU nano 6.2 process_manager.c
#define _XOPEN_SOURCE 700
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/time.h> // gettimeofday
#include <sys/wait.h> // waitpid, macros
#include <string.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h> // shmget, shmat
#include <signal.h>
#include <errno.h>

#define CHILD_COUNT 3
#define SHM_SIZE 4096

pid_t child_pids[CHILD_COUNT];
int pipes_arr[CHILD_COUNT][2];
int shmidx = -1;
int use_shm_for_child = -1; // which child uses shared memory (-1 = none)
volatile sig_atomic_t sigint_received = 0;
int nim_last_digit = 0;

void sigint_handler(int signo) {
    // Signal handler for SIGINT in parent (to gracefully terminate children).
    sigint_received = 1;
    // We DO NOT use non-async-signal-safe calls here except kill (which is safe).
    for (int i = 0; i < CHILD_COUNT; ++i) {
        if (child_pids[i] > 0) {
            kill(child_pids[i], SIGTERM); // request child termination
        }
    }
}

/* Utility: write full buffer to fd (handles short writes) */
ssize_t write_full(int fd, const void *buf, size_t count) {
    const char *p = buf;
    while (count > 0) {
        ssize_t r = write(fd, p, count);
        if (r < 0) return r;
        count -= r;
        p += r;
    }
    return count;
}
```

```
senjaramadhani@LAPTOP-82: ~$ nano process_manager.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -Wall -O2 -o process_manager process_manager.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./process_manager
=== Simple Process Manager (fork, pipe, shm, waitpid) ===
Masukkan digit terakhir NIM Anda (0-9): 3
NIM akhir GANJIL -> Child 2 akan menggunakan shared memory untuk komunikasi.
Input untuk Child 1 (faktorial): masukkan angka non-negatif (mis. 5): 3
Input untuk Child 2 (prima): masukkan batas (mis. 30): 43
Input untuk Child 3 (reverse string): masukkan string (max 1000 chars): 1043
Parent: created Child 1 with PID 405
Parent: created Child 2 with PID 406
Parent: created Child 3 with PID 407
Parent: Child index 1 finished. PID=405. Exited normally, exit status=0. Execution time (approx): 0.601 ms
Result from Child 1 via pipe:
Child 1 (PID 405): faktorial(3) = 6
Parent: Child index 2 finished. PID=406. Exited normally, exit status=0. Execution time (approx): 0.422 ms
Result from Child 2 via shared memory:
2 3 5 7 11 13 17 19 23 29 31 37 41 43
Parent: Child index 3 finished. PID=407. Exited normally, exit status=0. Execution time (approx): 0.223 ms
Result from Child 3 via pipe:
Child 3 (PID 407): reverse('1043') = '3401'
Parent: shared memory segment removed.
Parent: all children handled. Exiting.
senjaramadhani@LAPTOP-823QCJCF:~$
```

Tugas 2: File Monitor

```
GNU nano 6.2 file_monitor.c
/*
 * file_monitor.c (versi final)
 *
 * Program memantau direktori /tmp/monitor_dir
 * dan mencatat event IN_CREATE, IN_MODIFY, IN_DELETE.
 *
 * Tidak butuh argumen di command line.
 * Akan meminta digit terakhir NIM lewat input.
 */

#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/inotify.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <signal.h>
#include <string.h>
#include <time.h>
#include <errno.h>
#include <stdint.h>
#include <limits.h>

#ifndef NAME_MAX
#define NAME_MAX 255
#endif

#define BUF_LEN (10 * (sizeof(struct inotify_event) + NAME_MAX + 1))

static volatile sig_atomic_t running = 1;
static int inotify_fd = -1;
static int watch_fd = -1;
static int log_fd = -1;
static int nim_last_digit = 0;

[ Read 198 lines ]
Help      Write Out  Where Is    Cut         [ Read 198 lines ]  Location  M-U Undo  M-A Set Mark  M-] To Bracket
Exit      Read File   Replace    ^U Paste    ^J Justify  ^_/ Go To Line M-E Redo  M-C Copy  ^_ Where Was
```

```
int n = snprintf(logline, sizeof(logline),
                 "[%s] [%s] [%s] [%s]\n",
                 timestamp, evtype, name, pid_str);

if (should_log) write_all(log_fd, logline, (size_t)n);

if (!is_even) {
    printf("SIMULATED EMAIL: [%s] event=%s file=%s pid=%s\n",
           timestamp, evtype, name, pid_str);
}

i += sizeof(struct inotify_event) + ev->len;
}

printf("\nProgram dihentikan. Membersihkan resource...\n");
safe_close_fds();
return EXIT_SUCCESS;
}

senjaramadhani@LAPTOP-823QCJCF:~$ nano file_monitor.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -Wall -O2 -o file_monitor file_monitor.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./file_monitor
== File Monitor Sistem Operasi ==
Direktori default: /tmp/monitor_dir
Log file: /tmp/monitor.log
Masukkan digit terakhir NIM Anda: 3

Monitoring dimulai... (Ctrl+C untuk berhenti)
```

Tugas 3: Custom Shell

```
GNU nano 6.2 myshell.c
/*
 * Custom Shell - myshell.c
 * Description: Simple shell with manual parsing, background process, redirection, and multiple pipe support (for NIM with last digit odd)
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <fcntl.h>

#define MAX_CMD_LEN 1024
#define MAX_ARGS 128
#define MAX_PIPES 10

// Fungsi untuk membagi input menjadi argumen
void parse_command(char *cmd, char **args) {
    int i = 0;
    char *token = strtok(cmd, " \\t\\n");
    while (token != NULL) {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }
    args[i] = NULL;
}

// Fungsi untuk menjalankan perintah tunggal (tanpa pipe)
void execute_single_command(char **args, int background) {
    pid_t pid = fork();
    if (pid == 0) {
        if (execvp(args[0], args) == -1) {
            perror("myshell");
        }
        exit(EXIT_FAILURE);
    } else if (pid > 0) {
        // ...
    }
}
```

```
senjaramadhani@LAPTOP-82: ~$ nano myshell.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -Wall -O2 -o myshell myshell.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./myshell
Masukkan digit terakhir NIM: 3

myshell (NIM: 3, mode: MULTI-PIPE)
myshell$
```

Tugas 4: System Information Tool

```
GNU nano 6.2 sysinfo_tool.c
/*
 * System Information Tool
 * - Membaca file /proc/ menggunakan open() dan read()
 * - Menampilkan informasi CPU, Memory, Disk, Process, dan Network
 * - Tambahan: CPU load average atau TCP socket info tergantung NIM terakhir
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <dirent.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/statvfs.h>
#include <sys/types.h>
#include <ctype.h>
#include <errno.h>

#define BUF_SIZE 8192
#define PATH_MAX_LEN 512

int nin_last_digit;

/* Fungsi pembantu membaca isi file /proc menjadi string */
void read_file(const char *path, char *buffer, rsize_t size) {
    int fd = open(path, O_RDONLY);
    if (fd < 0) {
        perror("Error opening file");
        return;
    }
    ssize_t bytes = read(fd, buffer, size - 1);
    if (bytes > 0) buffer[bytes] = '\0';
    else buffer[0] = '\0';
    close(fd);
}

/* ===== CPU INFO ===== */
void show_cpu_info() {
    char data[BUF_SIZE];
    read_file("/proc/cpuinfo", data, sizeof(data));

    char model[128] = "N/A";
    int cores = 0;
    float freq = 0.0f;
}

[ Read 228 lines ]
[ Location: Go To Line ]
[ Undo: Redo ]
[ Set Mark: Copy ]
[ To Bracket: Where Was ]
[ Previous: Next ]
[ Back: Forward ]
[ Help: Exit ]
[ Write Out: Read File ]
[ Where Is: Replace ]
[ Cut: Paste ]
[ Execute: Justify ]
[ Hari panas yang... 26°C ]
[ Search ]
[ 23:53 16/10/2025 ]
```

```
senjaramadhani@LAPTOP-82: ~$ nano sysinfo_tool.c
senjaramadhani@LAPTOP-82: ~$ gcc -Wall -O2 -o sysinfo_tool sysinfo_tool.c
senjaramadhani@LAPTOP-82: ~$ ./sysinfo_tool
Masukkan digit terakhir NIM: 3

=== SYSTEM INFORMATION TOOL ===

===== CPU INFO =====
Model: AMD Ryzen 7 7435HS
Cores: 8 | Frequency: 3894.10 MHz

===== MEMORY INFO =====
Total: 3913924 kB | Used: 427784 kB | Free: 3333248 kB

===== DISK INFO =====
Total: 1031018 MB | Used: 3652 MB | Free: 1027365 MB

===== PROCESS LIST =====
| PID | Process Name | Status |
|-----|-----|-----|
1 | systemd | S
2 | init-systemd(Ub | S
7 | init | S
61 | systemd-journal | S
97 | systemd-udev | S
106 | systemd-resolve | S
110 | systemd-timesyn | S
106 | cron | S
188 | dbus-daemon | S
193 | my_daemon | S
194 | networkd-dispat | S
195 | rsyslogd | S
199 | systemd-logind | S
224 | sshd | S
225 | unattended-upgr | S
283 | login | S
287 | agetty | S
340 | systemd | S
341 | (sd-pam) | S
355 | bash | S
383 | agetty | S
574 | SessionLeader | S
575 | Relay(S80) | S
580 | bash | S
696 | sysinfo_tool | R

===== NETWORK INTERFACES =====
| Iface | RX Bytes | TX Bytes |
|-----|-----|-----|
[ Hari hujan yang... 26°C ]
[ Search ]
[ 23:52 16/10/2025 ]
```